

SOEN228/SOEN298 Lab Report Guidelines

Each student group is expected to submit one lab report per group submission, containing the following:

1. **A cover page with the following details:**

- Lab report number (Lab 0 of 3, Lab 1 of 3 ...etc)
- Team members' names and student ID numbers
- Date submitted
- Section

2. **Purpose of the lab:**

This section should include a brief description of the lab experiment and any necessary theory.

3. **Supporting diagram:** If the report requires a schematic, include a neat circuit diagram of your circuit used in the lab experiment. Use the appropriate logic symbols when necessary and label each pin and logic element using the following scheme: Unit/number/gate. If another diagram is necessary, please consult the lab demonstrator for more information. If the lab report requires a timing diagram, this would be the section to place it in.

Example: A student implements the following function with logic gates: $F(A,B,C) = \overline{(A \cdot B)} + C$. On the student's schematic, the gate implementing $A \cdot B$ is labeled U1A since it is the first unit (or chip) and gate A since it is implemented by using pins 1 to 3. The output is then inverted and the inverter is labeled U2A since it is the second chip used and only the first gate on the chip is used. To complete the final term, the student then uses the second AND gate on the 7408 IC with pins 4 to 6. They label the AND gate U1B. For more information, consult the SOEN228/SOEN298 Lab Manual.

Diagrams must be computer rendered and original work. Hand drawn schematics and photographs will not be accepted.

4. **Circuit operation:**

This section of the report should detail the operation of the circuit. The goal is to discuss how the circuits work from a functional standpoint. For the preliminary lab experiments, it is accepted to discuss how the circuit itself is connected. *Simply explaining in words how a circuit is wired will result in a poor mark since a schematic already provides this information.*

5. **Testing:**

Discuss how you tested the circuit and ensured its functionality. Go into detail about the inputs used to test and the corresponding outputs. Were these values expected or logical values? What went wrong and how did you resolve it?

Students are free to choose their tools for producing schematics and other supporting diagrams. That being said, below are some suggestions for tools to generate diagrams. Some may not have complete functionality to produce complete diagrams. It is up to the students to determine if the software they are using can yield proper diagrams. Additionally, it is the responsibility of the students to gain proficiency by self-study. The lab demonstrators are not responsible to teach students how to use the software they chose.

- Schematics:

1. KiCAD (Preferred, installed on all lab computers as well)
<http://kicad-pcb.org/>
2. SchemeIt by DigiKey (Web-based)
<http://www.digikey.ca/schemeit>

- Timing Diagrams:

1. WaveDROM (JSON-based generator, recommended)
<http://www.wavedrom.com>
2. TimingAnalyzer (Java-based generator)
<http://www.timing-diagrams.com>

Students are expected to adhere to the University's Code of Academic Conduct. Figures obtained from elsewhere must be properly referenced. Reliance on third party sources for content will be reflected in grades given.